

Phina Penteriani
2364350
Lab 5

1. The time-domain waveform doesn't show speech patterns very clearly. Two recordings of the same word can look pretty different as raw waveforms, especially if one is louder than the other. Converting to the frequency domain highlights the parts that actually matter for distinguishing words, like the harmonics and formants, so the model can focus on the content of the word instead of things like volume. This makes it much easier for a small model to learn and improves accuracy.
2. The representative dataset gives the converter real examples of what the model will see, so it can figure out the proper value ranges for each activation. INT8 quantization needs a scale and zero-point to map FP32 values into the INT8 range, and those can only be set correctly if the converter knows the actual range of activations. Without a representative dataset, the ranges would be guesses and the quantized model would lose accuracy.
3. `audio_processor.get_data()` loads the raw audio files from disk, turns them into spectrograms, and pairs them with their labels. It also adds some augmentation like random time shifts and background noise for the training set. The output is ready to be fed straight into the model.
4. `g_model` is the TFLite model stored as a byte array, and `g_model_len` is the length of that array. The interpreter on the Arduino reads `g_model` directly from flash to load the model, and uses `g_model_len` to know how big it is. If either one is wrong the model won't load correctly, so the array has to be copied into the sketch exactly as generated.

```
Heard unknown (207) @118050ms
Heard unknown (246) @118432ms
Heard no (203) @120048ms
Heard unknown (201) @124352ms
Heard unknown (210) @125872ms
Heard unknown (204) @128208ms
Heard unknown (228) @129744ms
Heard unknown (208) @131360ms
Heard unknown (213) @132880ms
Heard unknown (209) @134672ms
Heard unknown (236) @136192ms
Heard unknown (220) @137728ms
Heard unknown (210) @139792ms
Heard unknown (201) @142304ms
Heard unknown (202) @144720ms
Heard unknown (212) @147584ms
Heard no (201) @152624ms
Heard unknown (203) @158896ms
Heard unknown (203) @167600ms
```

5.